MIL-B-50230A(PA)
6 October 1975
SUPERSEDING
MIL-B-50230(PA)
20 March 1969

MILITARY SPECIFICATION

BATTERIES, STORAGE, 24 VOLTS AND 21 VOLTS

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for batteries, storage, 24 volts and 21 volts, used on MADM.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces For Organic Coatings.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	Environmental Test Methods
MIL-STD-1169	Packaging, Packing and Marking for Shipments of Inert Ammunition Components.

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DRAWINGS

PICATINNY ARSENAL

PRODUCT DRAWINGS

8808362	-	Packing and Marking for Storage and Shipments
8861672	-	Carton, Packing, Ammunition, for Battery, Storage
8862119	-	Battery, Storage, 24 Volts
8862120	-	Battery, Storage, 21 Volts
8883553	-	Carton, Packing, Ammunition, for Battery, Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.5) that all of the materials, parts, and components are in compliance with the applicable drawings.

3.1.1 Annealing. The annealing of cell cases shall be in accordance with the applicable drawing.

3.2 Assembly. The battery shall comply with all requirements and notes as specified on the drawings (8862119 and 8862120) with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.3).

3.4 Serialization. A serial number shall be assigned to each assembly by the contracting officer. The contracting officer shall request the serial code numbers from Picatinny Arsenal, SARPA-QA-N-A. The serial number shall be in addition to the normal lot numbers, but in no event shall there be any possibility of duplication nor any indication of order or size of production in either the lot designation, the serial number, or any other number.

3.5 Leakage. The individual cells of the battery assembly in the dry or activated state shall be capable of withstanding an internal pressure of 1.5 ± 0.5 pounds per square inch gage (p.s.i.g) for 30 seconds without evidence of leakage when tested as specified in 4.5.1.

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3.6 Insulation resistance, dry. The insulation resistance of the battery assembly, measured across the specified terminals, shall comply with the requirements specified in Table I, when tested as specified in 4.5.2.

Table I

Battery dwg. No.	Term (+)	to Term (-)	Megohms, min.
8862119	+	-	1.60
8862120	A	E	0.40
	E	D	0.20
	D	B	0.80
	B	A	1.40

3.7 Low temperature.

3.7.1 Storage. The battery assembly unactivated shall withstand exposure to a temperature of minus (-) 65 ± 5 degrees Fahrenheit (°F) for a period of 8 hours without evidence of cracks, embrittlement or other deterioration, when tested as specified in 4.5.3.1 and after test shall comply with 3.5 and 3.6.

3.7.2 Prior to Potting (Dwg. 8862120 only). - The battery circuit cells cemented together and connector straps tightened to required torque shall withstand exposure to a temperature of -25 ± 5°F and during test, shall comply with 3.16.1 or 3.16.2.2.1, (as applicable) when tested as specified in 4.5.3.3.

3.8 High temperature.

3.8.1 Battery, prior to potting (8862120 only). - The battery circuit cells cemented together and connector straps tightened to required torque shall withstand exposure to a temperature of +160 ± 5°F for a period of 4 hours without evidence of cracks, embrittlement or deterioration when tested as specified in 4.5.4.1 and after test shall comply with 3.5.

3.8.2 Storage. - The battery assembly unactivated shall withstand exposure to a temperature of +160 ± 5°F for a period of 8 hours without evidence of cracks, embrittlement or other deterioration, when tested as specified in 4.5.4.2 and after test shall comply with 3.5.

3.8.3 Activated. - The battery or batteries in the activated condition shall show no evidence of cracking, chipping, distortion or other physical deterioration, when subjected to a temperature of +125 ± 5°F and during test, shall comply with 3.16.1 or 3.16.2.2.2 (as applicable) when tested as specified in 4.5.4.3.

3.9 Activation. - The battery assembly fully charged shall produce the minimum voltage specified in Table II within 35 ± 5 minutes after the cells have been filled, when tested as specified in 4.5.5.

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Table II

Battery dwg. No.	Term. to Term.		Value D. C. Min. After act.
8862119	-	+	24 volts
8862120	A	E	-6.4 volts
	E	D	3.2 volts
	D	B	12.5 volts
	A	B	22.0 volts
	A	C	20.5 volts

3.10 Closed circuit voltage. - The battery assembly fully charged shall produce the minimum voltage specified in Table III while exposed to the applicable discharge rate, and the voltage reading shall be taken within 5 minutes after connecting load when tested as specified in 4.5.6.

Table III

Battery (Dwg 8862119) Terminal to terminal +	Discharge rate 200 ohms ± 10 ohms Value, d.c. 18 Volts, Min.
Battery (Dwg 8862120) Terminal to Terminal A B	Discharge rate 150 ohms ± 5 ohms Value d.c. 18 volts, min.

3.11 Impact shock. - The battery in the activated condition shall be rigidly secured to a test fixture and shall show no evidence of cracking, chipping, distortion or other physical deterioration when subjected to impact shocks of 15 g's, having a duration of 11 ± 1 milliseconds and after testing as specified in 4.5.7 shall comply with the requirements of 3.5 and 3.10.

3.12 Vibration. - The battery in the fully charged activated condition rigidly supported to a test fixture shall show no evidence of cracking, chipping, distortion or other physical deterioration when subjected to the vibrations specified in Table IV and when tested as specified in 4.5.8. After completion of the test, the battery shall comply with 3.5 and 3.10.

Table IV
Vibration Schedule

Frequency range Cycles per second	Amplitude
10-55-10	.06 inch
5-500-5	0.01 inch or 2gs max

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3.13 Thermal shock. - The battery in the activated condition shall withstand a thermal shock from +160 ± 5°F to -25 ±5°F, without evidence of cracking, chipping, distortion or other physical deterioration and after test, shall comply with the requirements of 3.5, when tested as specified in 4.5.9.

3.14 Altitude. - The battery in the activated condition shall withstand exposure to a pressure simulating an altitude of 30,000 ft. and shall be capable of meeting the requirements of 3.10 at a pressure simulating an altitude of 10,000 ft. without evidence of cracking, chipping, distortion or other physical deterioration and after test, shall comply with the requirements of 3.5, when tested as specified in 4.5.10.

3.15 Humidity. - The battery in the activated condition shall withstand exposure to a minimum relative humidity of 95 percent, without evidence of cracking, chipping, distortion, obliteration of marking or other physical deterioration and after test, shall comply with the requirements of 3.5, when tested as specified in 4.5.11.

3.16 Capacity.

3.16.1 Battery (Dwg 8862119). - The battery, fully charged shall be subjected to a fixed load of 200 ± 10 ohms and shall maintain a minimum potential of 18 v.d.c. for at least 90 minutes, when tested as specified in 4.5.12.1 and after test, shall comply with 3.5.

3.16.2 Battery (Dwg 8862120).

3.16.2.1 Low discharge rate. - The battery, fully charged shall be subjected to a fixed load of 150±5 ohms and shall maintain a minimum potential of 18 volts d.c. at -25°F and +160°F for a minimum of 48 hrs when tested as specified in 4.5.12.2.1 and shall comply with 3.5.

3.16.2.2 High discharge rate.

3.16.2.2.1 Low temperature. - A fully charged battery (see 3.16.2.2.3) shall be discharged at -25 ± 5°F through an adjustable resistor set for 60 (+5, -0) amperes. Without further adjustment of resistor, the battery shall be pulse discharged for 30 minutes at a rate of one pulse every 2 minutes. The pulse width shall be 2 ± .25 seconds duration. The pulse amplitude shall not drop below 45 amperes during the entire test period. The voltage between Terminal "A" and "E" measured across a fixed load of 5 amperes initial shall not go below 4.5 volts between pulses when tested as specified in 4.5.12.2.2 and after testing, the battery shall comply with 3.5.

3.16.2.2.2 High temperature. - A fully charged battery (see 3.16.2.2.3) shall be discharged at +125 ± 5°F through an adjustable resistor set for 150 ± 10 amperes. Without further adjustment the battery shall be pulse discharged for 30 minutes at a rate of one pulse every 2 minutes. The pulse width shall be 2±.25 seconds duration. The pulse amplitude

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shall not drop below 125 amperes during the entire test period. The voltage between Terminals "A" and "E" measured across a fixed load of 5 amperes initial shall not go below 5.0 volts between pulses when tested as specified in 4.5.12.2.2 and after testing the battery shall comply with 3.5.

3.16.2.2.3 Batteries. - Batteries for the first article test shall be tested for high discharge rate test within 7 days after activation.

3.17 Paint adhesion. - The painted surfaces of the battery shall show no evidence of peeling, flaking, chipping or other deterioration, when tested as specified in 4.5.13.

3.18 Polarity.

3.18.1 Single cells. - Single cells of batteries shall be checked for correct polarity when tested as specified in 4.5.14.

3.18.2 Monoblocks. - Battery monoblocks shall be checked for correct polarity of cells during assembly when tested as specified in 4.5.14.

3.19 Workmanship.

3.19.1 Plastic and metal characteristics. - All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.19.2 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.19.3 Burrs. - Parts shall be free from burrs.

3.19.4 Foreign matter. - Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.19.5 Cleaning. - The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 3.3)
2. Quality Conformance Inspection (see 4.4)

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4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Battery, Storage, 24 Volts	8862119	5
Battery, Storage, 21 Volts	8862120	5

4.3.2 Inspections to be performed. - The first article quantity submitted in accordance with 4.3.1 shall be subjected to any or all the requirements of the applicable drawings and to the inspections of Table V. Inspections or tests required in the Table V on parts and subassemblies before assembly shall be subject to verification by a representative of the Government designated by the Contracting Officer.

4.3.2.1 Battery, Storage, 21 Volts (Dwg. 8862120). - The contractor shall submit 5 battery assemblies, with cells properly cemented together and connector straps tightened to the required torque, prior to potting for tests 101, 131 and 132 of Table V. After successful completion of tests 101, 131 and 132, the battery assemblies shall be potted and the five (5) completed battery assemblies shall be subjected to the remaining first article examinations or tests 133 through 148 of Table V.

4.3.2.2 Battery, Storage, 24 Volts (Dwg. 8862119). - The contractor shall submit five (5) complete battery assemblies for the first article examinations or tests of Table V. After successful completion of tests 101, 131 and 132, the battery assemblies shall be potted and the five (5) completed battery assemblies shall be subjected to the remaining first article examinations or tests 133 through 148 of Table V.

4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination.- See tables in Para 4.3.2.1, 4.3.2.2, 4.3.2.3 and MIL-A-48078.

a. Major A and B defects sampling plans. - Unless otherwise specified, inspection for Major A and Major B defects shall be sampled in accordance with Table VI.

b. Minor defects sampling plans. - Unless otherwise specified, inspection for minor defects shall be sampled in accordance with Table VII.

c. Production tests sampling plans.- Unless otherwise specified, production tests shall be sampled in accordance with the tables specified in the Quality Conformance Inspection, Classification of Defects and Tests sheets.

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TABLE V

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.2.2.1	Battery, Storage, 24 Volts Battery, Storage, 21 Volts				8862119 8862120
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Polarity of single cell and polarity of monoblock	5	100%	3.18	4.5.14
Major B					
131	Low temperature (before potting, non-destructive)	5	100%	3.7.2	4.5.3.2
132	High temperature (before potting, non-destructive)	5	100%	3.8.1	4.5.4.1
133	Classification of characteristics (after potting, non-destructive)	5	100%	4.3.2.1	4.5
134	Leakage (after potting, non-destructive)	5	100%	3.5	4.5.1
135	Insulation resistance (after potting, non-destructive)	5	100%	3.6	4.5.2
136	Low temperature (after potting, non-destructive)	5	100%	3.7.1	4.5.3.1
137	High temperature (after potting, non-destructive)	5	100%	3.8.2	4.5.4.2
138	Activation (after potting, destructive)	5	100%	3.9	4.5.5
139	Closed circuit voltage (after potting, destructive)	5	100%	3.10	4.5.6

NOTES:

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TABLE V (Continued)

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.2.2.1	Battery, Storage, 24 Volts Battery, Storage, 21 Volts				8862119 8862120 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
140	Thermal shock (after potting, destructive)	5	100%	3.13	4.5.9
141	Altitude (after potting, destructive)	5	100%	3.14	4.5.10
142	Humidity (after potting, destructive)	5	100%	3.15	4.5.11
143	Paint adhesion (after potting, non-destructive)	5	100%	3.17	4.5.13
144	Low temperature (after potting, destructive)	5	100%	3.7.3	4.5.3.3
145	High temperature (after potting, destructive)	1*		3.8.3	4.5.4.3
146	Impact shock (after potting, destructive)	1*		3.11	4.5.7
147	Vibration (after potting, destructive)	1*		3.12	4.5.8
148	Capacity (after potting, destructive)	2**		3.7.3	4.5.3.3
	Battery (8862119)			3.8.3	4.5.4.3
	Battery (8862120)			3.16.2.1	4.5.12.2.1

NOTES: * One fully charged battery selected at random from the first article sample.
** One fully charged battery is selected for the low temperature test, and the other fully charged battery for the high temperature test.

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TABLE VI
Major Defects

LQ = .01 Acceptance No = 0 Consumers Risk = .05

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	97	93	126-127	115	162-163	137
73	72	98	94	128	116	164-165	138
74	73	99-100	95	129-130	117	166-167	139
75	74	101	96	131	118	168	140
76	75	102	97	132-133	119	169-170	141
77	76	103	98	134	120	171-172	142
78	77	104-105	99	135-136	121	173-174	143
79	78	106	100	137	122	175-176	144
80	79	107	101	138-139	123	177-178	145
81-82	80	108-109	102	140-141	124	179-180	146
83	81	110	103	142	125	181-182	147
84	82	111	104	143-144	126	183-184	148
85	83	112-113	105	145	127	185-187	149
86	84	114	106	146-147	128	188-189	150
87	85	115	107	148-149	129	190-191	151
88-89	86	116-117	108	150	130	192-193	152
90	87	118	109	151-152	131	194-195	153
91	88	119	110	153-154	132	196-197	154
92	89	120-121	111	155-156	133	198-200	155
93	90	122	112	157	134		
94-95	91	123-124	113	158-159	135		
96	92	125	114	160-161	136		

TABLE VII
Minor Defects

LQ = .05 Acceptance No = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.3.2.1	Battery, Storage, 24 Volts Battery, Storage, 21 Volts		8862119 8862120
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Polarity #1 single cell, polarity #2 monoblock not correct		Table VI	3.18	Visual
Major B					
131	Material not correct		-	3.1	Visual
132	Objective evidence (see 6.5) available for annealing in accordance with applicable drawing.		Table VI	3.1.1	Visual/Manual
133	Cells damaged or improperly joined before potting.		Table VI	3.2	Visual
134	Terminals improperly connected before potting.		Table VI	3.2	Visual
135	Improper torque (8862120 only) before potting.		Table VI	3.2	Gage
136	Insufficient bonding (cells and cover not completely bonded) before potting.		Table VI	3.19	Visual
137	Insufficient cement between cells before potting.		Table VI	3.2	Visual
138	Radius at filling holes, min		Table VI	3.2	Gage
139	Total length		Table VI	3.2	Gage
140	Total width		Table VI	3.2	Gage
141	Height		Table VI	3.2	Gage
142	Location of terminals		Table VI	3.2	Gage
143	Terminal damaged		Table VI	3.19	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 8862119 8862120
4.3.2.1	Battery, Storage, 24 Volts Battery, Storage, 21 Volts				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
144	Nut, hexagon, loose, missing or damaged before potting.		Table VI	3.2	Visual/Manual
145	Caps missing or damaged		Table VI	3.2	Visual
146	Sponge covering each cell missing		Table VI	3.2	Visual
147	Vent in cap missing or plugged		Table VI	3.2	Visual
148	Cracks or voids in potting compound or in battery		Table VI	3.19	Visual
149	Objective evidence (see 6.5) available for formulation (color, concentration, hardness, method of application) of potting compound.		Table VI	3.1	Visual/Manual
Minor					
201	Marking missing, improper or illegible		Table VII	5.2.3	Visual
202	Evidence of poor workmanship		Table VII	3.19	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.2	Carton, Packing, Ammunition, for Battery, Storage	1 OF 1	8861672 8883553
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major B					
131	Packaging, packing and marking	5	Table VI	5.0	MIL-STD-1169
132	Objective evidence for packaging materials	5	Table VI	6.2	Visual
133	Contents loose in carton	5	Table VI	3.2	Manual
134	Contents improperly sealed	5	Table VI	3.2	Visual
135	Label not present	5	Table VI	3.2	Visual
136	Box corners not blunted	5	Table VI	3.2	Visual
137	Air not exhausted from bag	5	Table VI	3.2	Visual
138	Objective evidence for heat seal test	5	Table VI	6.2	MIL-P-116
Minor					
201	Marking not present, incorrect or illegible	5	Table VII	5.2.3	Visual
202	Evidence of poor workmanship	5	Table VII	3.19	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.3	Packing and Marking for Storage and Shipments				8808362
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Gross load weight, max	5	Table VI		Measure *
132	Contents not tight in package	5	Table VI		Manual
133	Incorrectly packed	5	Table VI		Visual/Manual
134	Package material improper	5	Table VI	3.1	Visual
Minor					
201	Marking illegible, incomplete, absent	5	Table VII	5.2.3	Visual
202	Workmanship	5	Table VII	3.19	Visual/Manual

NOTES:

* Determine weight with a suitable scale.

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4.4.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests shall be performed in the order specified.

	Test	Before Potting	After Potting	Before Activation	After Activation
a.	Polarity (3.18)	X		X	
b.	Low Temperature (3.7.2)	X		X	
c.	High Temperature (3.8.1)	X		X	
d.	Leakage (3.5)	X		X	
e.	Insulation Resistance (3.6)	X		X	
f.	Low Temperature (3.7.1)		X	X	
g.	High Temperature (3.8.2)		X	X	
h.	Leakage (3.5)		X	X	
i.	Insulation Resistance (3.6)		X	X	
j.	Activation (3.9)		X		
k.	Closed Circuit Voltage (3.10)		X		X
l.	Low Temperature (3.7.3)		X		X
m.	High Temperature (3.8.3)		X		X
n.	Capacity (3.16.1 or 3.16.2.1)		X		X
o.	Paint adhesion (3.17)		X		X

Of these tests, j,k,l,m and n are considered destructive. Samples so tested shall not be returned to the lot. The destructive samples are over and above the deliverable quantity of batteries required by the contract.

4.4.3.1 Low Temperature (see 3.7.2), High Temperature (3.8.1), Leakage (3.5), Paint Adhesion (3.17). - Each battery in the lot shall be subjected to these tests in the order specified (see applicable paragraph in 4.5). If any battery fails to comply with the applicable requirements, it shall be classified defective and removed from the lot.

4.4.3.2 Low Temperature (see 3.7.1), High Temperature (3.8.2), Leakage (3.5), Insulation resistance (3.6), Activation (3.9), Closed Circuit Voltage (3.10), Low Temperature (3.7.3), High Temperature (3.8.3), Capacity (3.16.1 or 3.16.2.1). - One battery out of every 36 or less batteries that are produced shall be subjected to these tests in the order specified (see applicable paragraph in 4.5). If the battery fails to comply with the applicable requirements, the lot shall be rejected.

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4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.2.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

4.5 Test methods. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.5.1 Leakage (see 3.5). - Each individual cell of the battery shall be tested for leakage utilizing an approved test fixture. The fixture shall limit the pressure for a maximum of 3 p.s.i.g. for personnel safety and the pressure shall be applied using the applicable nozzle. The pressure supply shall be cut off and the cell shall maintain the internal pressure for not less than 30 seconds as determined by monitoring the gage. No adjoining cells shall be tested at the same time. Test is performed on dry and activated batteries.

4.5.2 Insulation resistance, dry (see 3.6). - The insulation resistance shall be measured with approved insulation resistance measuring equipment and subjected to a test potential of 500±50 volts direct current. Test is performed on dry batteries only.

4.5.3 Low Temperature.(see 3.7).

4.5.3.1 Storage (see 3.7.1). - The dry battery shall be placed in a chamber and the temperature shall be reduced to -65±5°F. Upon stabilization (see 6.4), the battery shall be exposed to the temperature of -65°F for a minimum period of 8 hours. At the end of this time, the battery shall be removed and brought to ambient temperature. The battery shall comply with 3.5 and 3.6.

4.5.3.2 Prior to potting (8862120 only) (see 3.7.2). - The unpotted battery shall be placed in a chamber and the temperature of the chamber shall be reduced to -65±5°F and stabilized (see 6.4). After stabilization, the battery shall be exposed to the temperature of -65°F for a period of 4 hours. At the end of this time, the battery shall be removed from the chamber, and examined for deterioration and damage.

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4.5.3.3 Activated (see 3.7.3). - The activated battery assembled in the respective battery case shall be placed in a test chamber and the temperature shall be lowered to -25±5°F. Upon stabilization (see 6.4) the battery or batteries shall be exposed to the temperature for a minimum period of 8 hours. At the end of this time and with the temperature of the chamber maintained at -25±5°F, the battery shall comply with 3.16.1 or 3.16.2.2.1 as applicable.

4.5.4 High temperature (see 3.8).

4.5.4.1 Prior to potting (see 3.8.1).- The unpotted battery shall be placed in a chamber and the temperature of the chamber shall be raised to +160±5°F and stabilized (see 6.4). After stabilization, the battery shall be exposed to the temperature of +160°F for a period of 4 hours. At the end of this time, the battery shall be removed from the chamber and shall comply with 3.5.

4.5.4.2 Storage (see 3.8.2). - The dry battery shall be placed in a chamber and the temperature raised to +160±5°F for a minimum period of 8 hours. At the end of this time the battery shall be removed, brought down to ambient temperature, examined thoroughly for deterioration and damage, and shall comply with 3.5

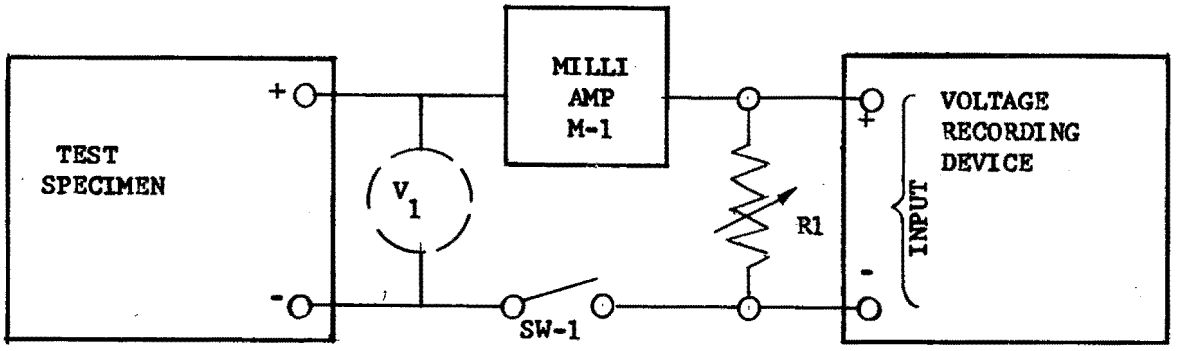
4.5.4.3 Activated (see 3.8.3). - The activated battery assembled in its respective battery case shall be placed in a test chamber and the temperature shall be raised to +125 ±5°F. Upon stabilization (see 6.4), the battery or batteries shall be exposed to the temperature of +125°F for a minimum period of 8 hours. At the end of this time and with the temperature of the chamber maintained at +125±5°F the battery shall comply with 3.16.1 or 3.16.2.2.2 as applicable.

4.5.5 Activation (see 3.9). - The battery inserted into a battery box shall be activated. The voltage shall be determined 35 ± 5 minutes after the last cell has been filled, utilizing approved measuring equipment. Proper polarity shall be observed in determining voltage.

NOTE: Swelling of the cells upon activation is not to be considered detrimental.

4.5.6 Closed circuit voltage (see 3.10).- The closed circuit voltage of the battery shall be determined by utilizing the test circuit shown on figure 1. The load shall be adjusted for the applicable discharge rate, as measured on meter (M-1). The voltage shall be measured on voltmeter (V-1).

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NOTE 1: MILLIAMMETER M-1 IS SHORTED OUT AFTER RESISTOR R1 HAS BEEN ADJUSTED FOR THE DESIRED LOAD.

FIGURE 1

4.5.7 Impact shock (see 3.11). - The battery, activated and rigidly mounted into an approved test fixture, shall be subjected to a total of 18 impact shocks, three shocks in each direction of its three mutually perpendicular axes. Each shock shall be a minimum of 15 gravity units (g's) with a time duration of 11 ± 1 milliseconds. After completion of the shock test, the battery shall comply with 3.5 and 3.10 and examined thoroughly for deterioration and damage.

4.5.8 Vibration (see 3.12). - The battery or batteries, activated and rigidly mounted into an approved test fixture, shall be vibrated through the frequency range of 5-500-5 c.p.s. with an applied vibratory acceleration of 2 g's or a constant double amplitude of 0.01 inches (total excursion). The range shall be traversed in 30 minutes and the total test time in each plane shall be 2 hours. Each of the three mutually perpendicular planes shall be vibrated in this manner. After completion of the high frequency vibration test, the battery shall

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be vibrated through the frequency range of 10-55-10 cycles per second with a constant double amplitude of 0.06 inches (total excursion). The range shall be traversed in 2 minutes and the total test time in each plane shall be 45 minutes. Each of the three mutually perpendicular planes shall be vibrated in this manner. After completion of the test, the battery shall comply with 3.5 and 3.10 and examined thoroughly for deterioration and damage.

4.5.9 Thermal shock. - Two temperature chambers shall be utilized for this test. Upon stabilization (see 6.4) of one chamber at +160±5°F, the activated battery shall be placed in this chamber and the chamber shall again be stabilized at +160°F. The exposure time to the +160°F temperature shall be a minimum of 4 hours. The temperature of the second chamber shall be stabilized at -25 ± 5°F and at the end of the 4 hour period at +160°F, the battery or batteries shall be transferred from the first chamber to the second chamber. This transfer shall be without delay. The second chamber shall again be stabilized. After stabilization, the exposure time to the -25°F temperature shall be a minimum of 4 hours. At the end of this time, the battery or batteries shall be removed, examined thoroughly for deterioration and damage, and shall comply with 3.5.

4.5.10 Altitude (see 3.14). - The activated battery shall be placed in a test chamber, the temperature of which has stabilized (see 6.4) at -25 ± 5°F. The temperature of -25°F shall be maintained throughout this test. The pressure shall be reduced within 30 minutes, to that pressure simulating a minimum altitude of 30,000 feet. The pressure shall be maintained for a minimum of 1 hour. At the end of this time, the pressure shall be raised to the ambient pressure and the battery or batteries shall be removed, examined thoroughly for deterioration and damage, and shall comply with 3.5. When testing facilities permit, this test can be run simultaneously with the test in 4.4.4.3.

4.5.11 Humidity. - The activated battery shall be placed on its side in a test chamber in such a manner that any accumulated moisture will run away from the terminals being used for electrical connection. The chamber temperature shall be raised to +125±5°F within the first 2 hours and maintained at this temperature for 6 hours. The temperature shall be reduced over the next 16 hour period to +90±5°F. The relative humidity throughout the 24 hour cycle shall be a minimum of 95 percent. The battery or batteries shall be subjected to three such cycles without interruption. At the end of the 72 hour period, they shall be removed from the chamber. The excess moisture shall be wiped off. The battery or batteries shall be examined thoroughly for deterioration and damage, and shall comply with 3.5.

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4.5.12 Capacity (see 3.16).

4.5.12.1 Battery (Dwg. 8862119)(See 3.16.1). - The battery shall be connected to an approved Rheostat and an approved recording device. Utilizing a bridge network, adjust the Rheostat for a load of 200±10 ohms. The voltage shall be recorded continuously and although the test requirement specifies a minimum of 90 minutes, testing shall be continued until the voltage falls below 18 v.d.c. Upon completion, the battery shall comply with 3.5.

NOTE: During temperature tests, there shall be no drain on the battery until the required soak time is completed.

4.5.12.2 Battery (Dwg. 8862120) (See 3.16.2).

4.5.12.2.1 Low discharge rate. - One (1) fully charged battery shall be placed in a test chamber and the temperature lowered to -25±5°F for a minimum period of eight (8) hours. The fully charged battery shall then be connected across terminals A(-) and B(+) at the specified load to an approved recording device. The voltage shall be recorded continuously and although the test requirement specifies a minimum of 48 hours, testing shall be continued until the voltage falls below 18 V.D.C. On completion, the battery shall comply with 3.5. One (1) fully charged battery shall be placed in a test chamber and the temperature raised to +160±5°F for a minimum period of eight (8) hours. The fully charged battery shall then be connected across terminals A(-) and B(+) at the specified load to an approved recording device. The voltage shall be recorded continuously and although the test requirement specifies a minimum of 48 hours, testing shall be continued until the voltage falls below 18 V.D.C. On completion, the battery shall comply with 3.5.

NOTE: During temperature tests, there shall be no drain on the battery until the required soak time is completed.

4.5.12.2.2 High discharge rate (see 3.16.2.2). - The battery shall be connected to the load specified, with the voltage continuously monitored utilizing an approved Voltmeter. The voltage measured from terminal "A" and terminal "E" shall not drop below the applicable d.c. voltage as specified in section 3. Upon completion, the batteries shall comply with 3.5.

NOTE: During temperature tests, there shall be no drain on the batteries until required soak time is completed.

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4.5.13 Paint adhesion (see 3.17). - The test shall be conducted in accordance with TT-C-490.

4.5.14 Polarity (see 3.18). - Single cells and cells assembled in monoblocks shall be tested for correct polarity utilizing approved inspection equipment to be furnished by the contractor.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1.1 Level A. - The battery shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.1.2 Level C. - The battery shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Packing.

5.1.2.1 Levels A and C.- The battery shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.2 Repair Parts.

5.2.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1.1 Level A. - The battery shall be packaged in accordance with Dwgs 8861672 or 8883553 as applicable and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. - The battery shall be packaged in accordance with Dwgs 8861672 or 8883553 as applicable and 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C.- The battery shall be packed in accordance with Dwgs 8861672 or 8883553 as applicable and 8808362.

6. NOTES

6.1 Intended use.- The batteries covered by this specification are intended for use on the Medium Atomic Demolition Munition (MADM).

6.2 Ordering data. - See MIL-A-48078

6.3 Submission of Designs for Approval.- See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

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6.4 Temperature stabilization. - The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min., do not exceed the nominal temperature plus tolerance specified.

6.5 Objective evidence. - Records of contractor quality control which can be verified.

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